

Assignment-2: Question-Answer

- **Explain what is serverless architecture?**

Serverless architecture is a cloud computing model where the developer does not need to manage the servers directly. The servers still exist, but the cloud provider takes care of the infrastructure, maintenance and scaling. The developer mainly writes and deploys the application code or functions.

In my Week-2 project, I used CSC Rahti to deploy a web application inside a Docker container. I provided my HTML file and Dockerfile, and Rahti/OpenShift created the build, deployment, pod, service and route needed to run my application. I did not have to create or maintain the physical server.

- **When it is more feasible solution to use application containers (app virtualization) than full OS virtualization to provide networked services?**

Application containers are more feasible than full OS virtualization when we need to run small or separate networked services without creating a complete virtual machine for each service. Containers are usually lighter and faster to start, and they make it easier to deploy and scale applications.

In my Week-2 project, I used a Docker container on CSC Rahti to run an Apache web application. I provided my HTML file and Dockerfile, and Rahti/OpenShift handled the deployment. In comparison, in my Week 1 project I created a complete Ubuntu VPS and installed Apache manually. Therefore, for a simple web service like my Week 2 project, a container is more convenient than creating a complete virtual machine.

- **List some information security risks when using third party (for example Docker) containers?**

Using third-party Docker containers can create several information security risks. The container may contain vulnerable or outdated software, or it could come from an untrusted source and contain malicious code. It can also be difficult to know all the packages and dependencies inside the container. Another risk is giving the container more permissions than it needs. If the container is compromised, the attacker may be able to access data or other resources. Therefore, it is important to use trusted container images, keep them updated and use only the permissions that are necessary.

In my project, the idea was:

Dockerfile → Apache container image → My webpage → Rahti

I did not build every part of Apache myself. Therefore, I need to trust and maintain the software/components used by the container.

This is why it is important to use images from trusted sources, keep them updated, and avoid giving containers unnecessary permissions.

- **If you would now use a virtualized Windows operating system in your laptop (whatever operating system the laptop already has): Which virtualization platform (VMware, Virtualbox, Hyper-V, KVM, etc...) would you use and why?**

If I wanted to use a virtualized Windows operating system on my laptop, I would use VirtualBox. I would choose VirtualBox because it is easy to install and use, and it allows me to create and run a virtual machine without changing the operating system already installed on my laptop. It is also useful for learning and testing because I can create a separate Windows environment and use it without affecting my main system.

For example, my laptop could continue running its normal Windows operating system, while VirtualBox runs another Windows operating system inside a virtual machine. This would allow me to test software or practice different configurations safely in the virtual machine.